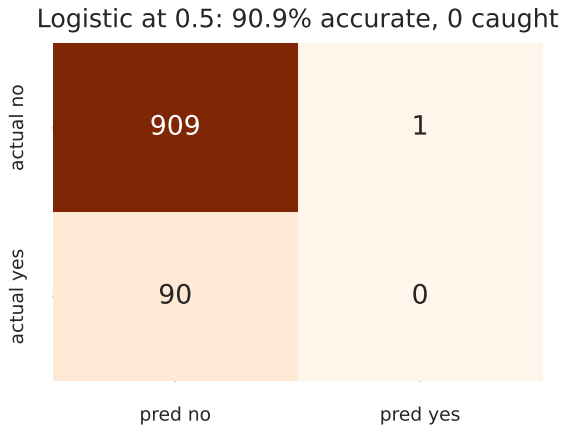
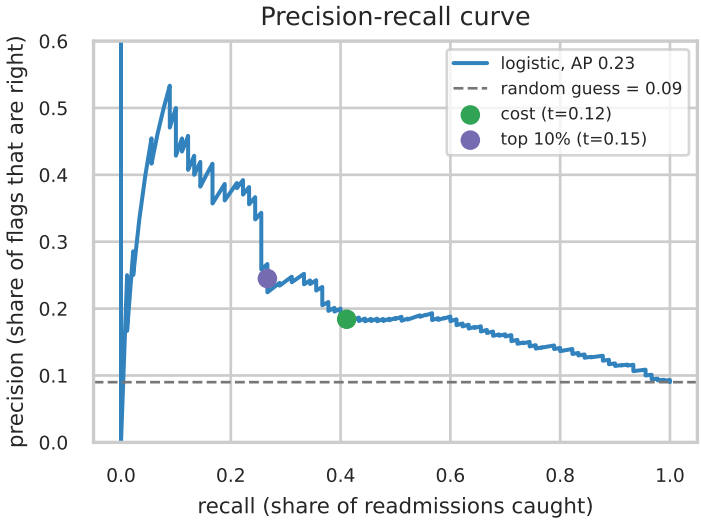
Day 18 Cheat Sheet: Imbalanced Outcomes and Thresholds

Week 3: accuracy trap, precision-recall, cost and capacity thresholds on 30-day readmissions

TERM	WHAT IT MEANS	CODE
Imbalanced	the outcome is rare (9% readmitted here)	y.mean()
Accuracy trap	"nobody" model scores 91%; accuracy is useless here	accuracy_score
Precision	of patients flagged, share truly readmitted	precision_score
Recall (sensitivity)	of readmitted patients, share we flagged	recall_score
PR curve / AP	precision vs recall at every threshold; baseline=rate	average_precision_score
Threshold	flag if risk >= t; .predict() silently uses 0.5	(p >= t).astype(int)
Out-of-fold scores	CV predictions on train, used to choose t	cross_val_predict
Break-even risk	call if p * benefit > cost of the call	cost / (prevent*$)
class_weight	upweights rare class; same ranking, inflated risks	class_weight="balanced"
Capacity rule	flag the top k% because staff time is fixed	np.quantile(oof, .9)





THIS DATA (same simulated days 14-17 data, seed 14): 4,000 discharges, 9.0% readmitted, 3,000 train / 1,000 test
Logistic at default 0.5: flags 1 patient, catches 0 of 90 readmissions, still 90.9% accurate
ROC AUC 0.723 but average precision 0.230 (random guess 0.090): about 2.5x better than chance
Cost rule (illustrative $15k readmit, $500 call, 25% prevented): t = 0.12, flags 201, precision 0.18, recall 0.41
Capacity rule (top 10%): t = 0.154, flags 98, precision 0.24, recall 0.27
class_weight='balanced': AUC 0.724 (same), mean risk 0.47 vs true 0.09, Brier 0.226 vs 0.077

HABITS THAT KEEP YOU HONEST
- Never report accuracy alone for a rare outcome; give precision, recall and the base rate
- Choose the threshold on training data (out-of-fold), then apply it once to test
- The threshold is a business decision: cost of a miss, cost of a call, staff capacity
- Reweighting or resampling changes probabilities, not ranking; recalibrate before sharing risks
- Report PR AUC next to ROC AUC; the PR baseline is the prevalence, not 0.5

```
from sklearn.model_selection import cross_val_predict
oof = cross_val_predict(LogisticRegression(max_iter=1000), X_train, y_train,
                        cv=5, method="predict_proba")[:, 1]
t = np.quantile(oof, 0.90)          # flag the riskiest 10%
flag = (model.predict_proba(X_test)[:, 1] >= t).astype(int)
```

Infographic: 91% Accurate, 0% Useful

Same simulated readmissions as days 14-17. The threshold is where the model meets the care team

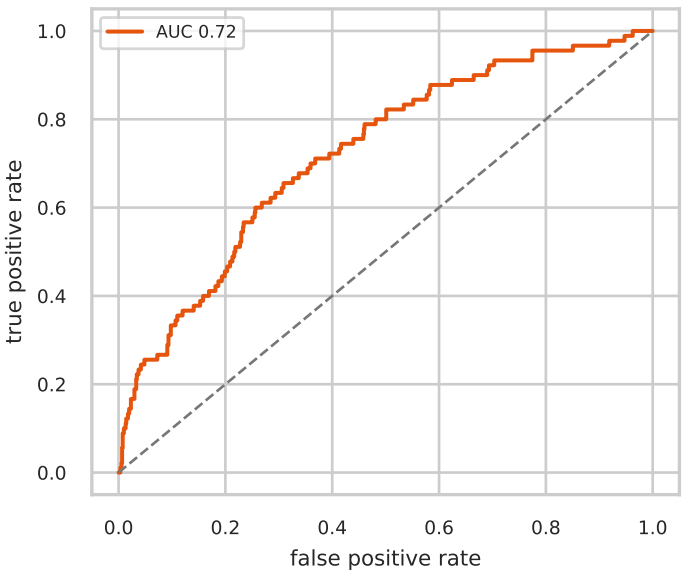
DEFAULT 0.5
91% accurate
catches 0 of 90

PR AUC
0.23 vs 0.09
random guess

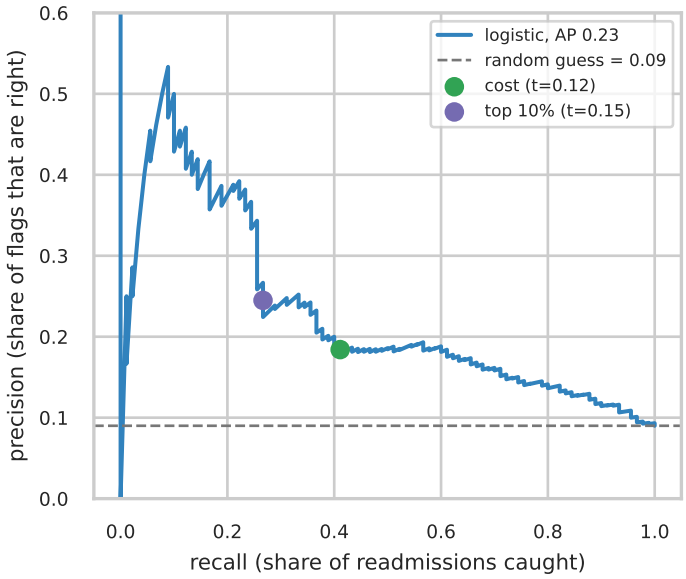
COST RULE
t=0.12: flags 201
catches 37 of 90

TOP 10%
t=0.15: flags 98
catches 24 of 90

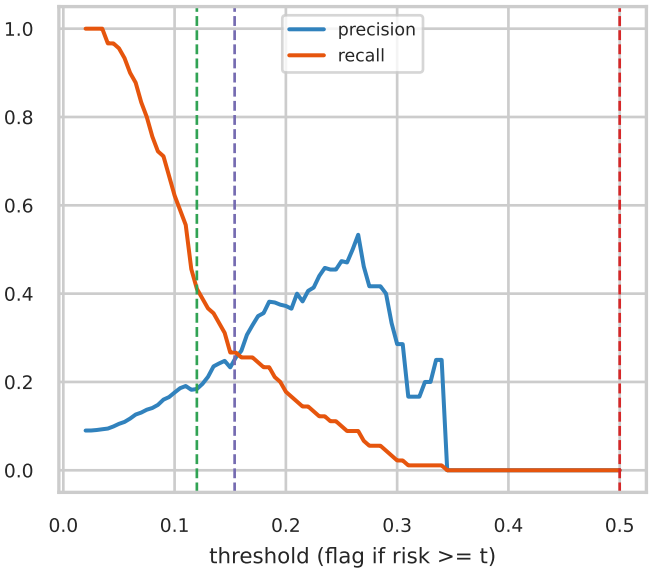
ROC looks fine; hides the rarity



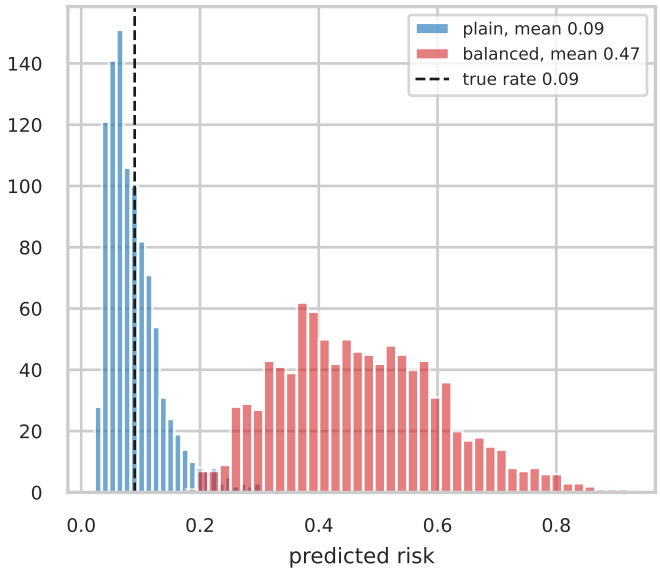
Precision-recall curve



Moving the threshold trades recall for precision



class_weight='balanced' inflates risk



Takeaways
- Accuracy rewards ignoring the rare outcome; use precision and recall
- The PR curve's floor is the readmission rate (0.09), so AP 0.23 is a real but modest signal
- Pick the threshold from costs or staff capacity, using training data only
- Balancing classes moves the threshold for you but breaks probabilities
- Tomorrow: unsupervised learning, grouping patients with k-means clustering

Hands-On Worksheet: ds-genai-90-days / day18

60-90 minutes | notebook: day18/thresholds.ipynb + day18/README.md

[ ] mkdir day18, conda activate ds90, open day18/thresholds.ipynb

[ ] Rebuild the days 14-17 data and the same split (snippet 1)

[ ] Accuracy trap: score the 'nobody' model, then logistic at 0.5 (snippet 2)

[ ] Print and label the confusion matrix (rows actual, cols predicted)

[ ] PR curve + average precision; draw the prevalence baseline (snippet 3)

[ ] Table of precision / recall / flags at 5 thresholds

[ ] Out-of-fold scores on TRAIN with cross_val_predict (snippet 4)

[ ] Cost curve: find the cheapest threshold; check the break-even math

[ ] Capacity rule: threshold for the top 10%; apply both rules to test once

[ ] class_weight='balanced': compare AUC, AP, mean risk, Brier (snippet 5)

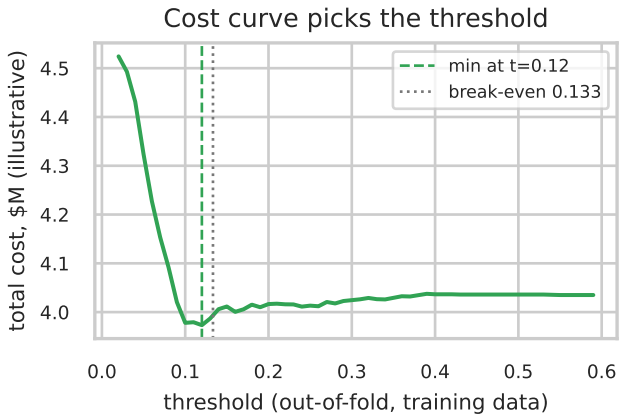
[ ] Plot plain vs balanced risk histograms with the true rate

[ ] README: 3 charts, a threshold table, and a recommended rule with reasons

[ ] git add day18 && git commit && git push (snippet 6)

[ ] Stretch: precision / recall by threshold in Snowflake SQL (snippet 7)

```
prec, rec, thr = precision_recall_curve(
    y_test, p_lr)
ap = average_precision_score(y_test, p_lr)
flag = (p_lr >= t).astype(int)
precision_score(y_test, flag)
recall_score(y_test, flag)
```



Accuracy at 0.5: _____ ROC AUC: _____ AP: _____ Prevalence: _____

Cost t: ____ flags/recall ____/____ Top-10% t: ____ flags/recall ____/____ Balanced mean risk: ____

Day 18 Self-Check Quiz

1. Why is 91% accuracy meaningless for a 9% readmission outcome?

   _____

2. Define precision and recall in plain words for a readmission program.

   _____

3. Why does logistic regression flag almost nobody at the default 0.5?

   _____

4. What is the random-guess baseline for average precision? For ROC AUC?

   _____

5. Why choose the threshold with out-of-fold training scores, not the test set?

   _____

6. A call costs $500, prevents 25% of $15k readmissions. Break-even risk?

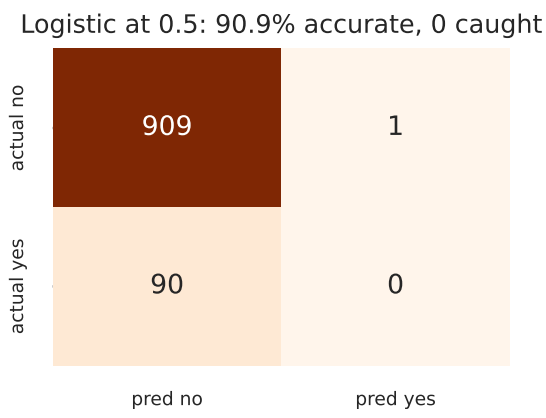
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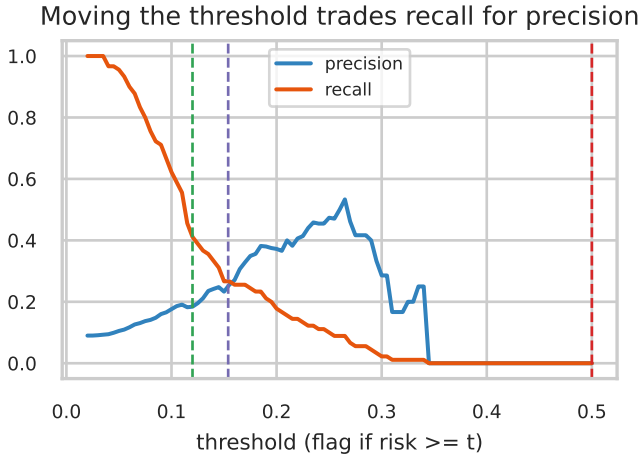
7. class_weight='balanced' kept AUC but mean risk became 0.47. What happened?

   _____

8. Staff can call 100 patients a week. Which rule fits, and what do you report?

   _____





Answers:
1 predicting 'nobody is readmitted' already scores 91%; accuracy ignores whether you catch the rare cases you care about
2 precision: of patients we flag, share truly readmitted; recall: of patients readmitted, share we flagged in advance
3 almost no patient has predicted risk above 50% when the base rate is 9%; the default threshold assumes balanced classes
4 AP baseline = prevalence (0.09); ROC AUC baseline = 0.5. That is why AP 0.23 looks low but is 2.5x chance
5 the test set must stay untouched; choosing t on it leaks information and makes test precision / recall optimistic
6 call when 0.25 x p x 15,000 > 500, so p > 500 / 3,750 = 0.133 (the curve's minimum, 0.12, is close)
7 it upweighted readmissions, which shifts every risk up: same ranking, so the same AUC, but risks are no longer real probabilities
8 the capacity rule (top ~10%); report the threshold, flags, precision and recall, e.g. 98 flagged, 24 of 90 readmissions caught